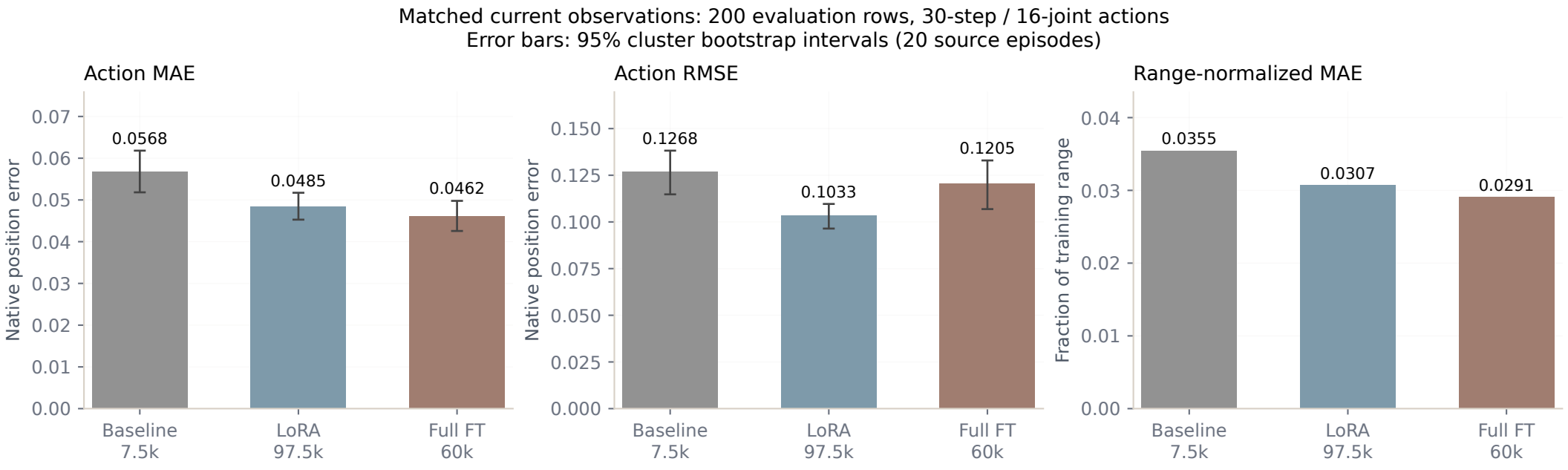
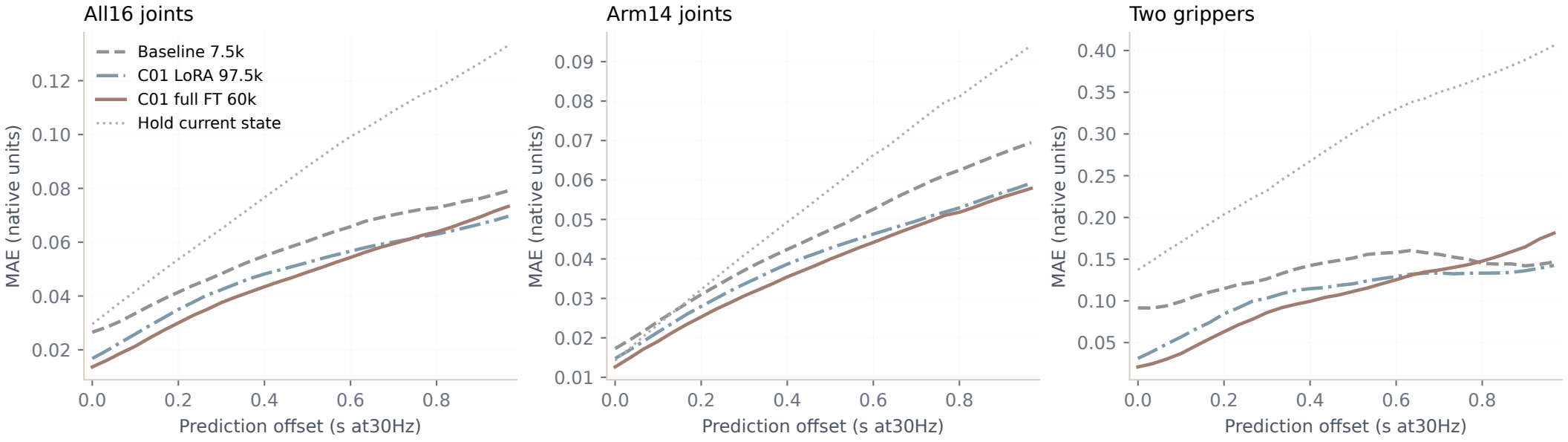
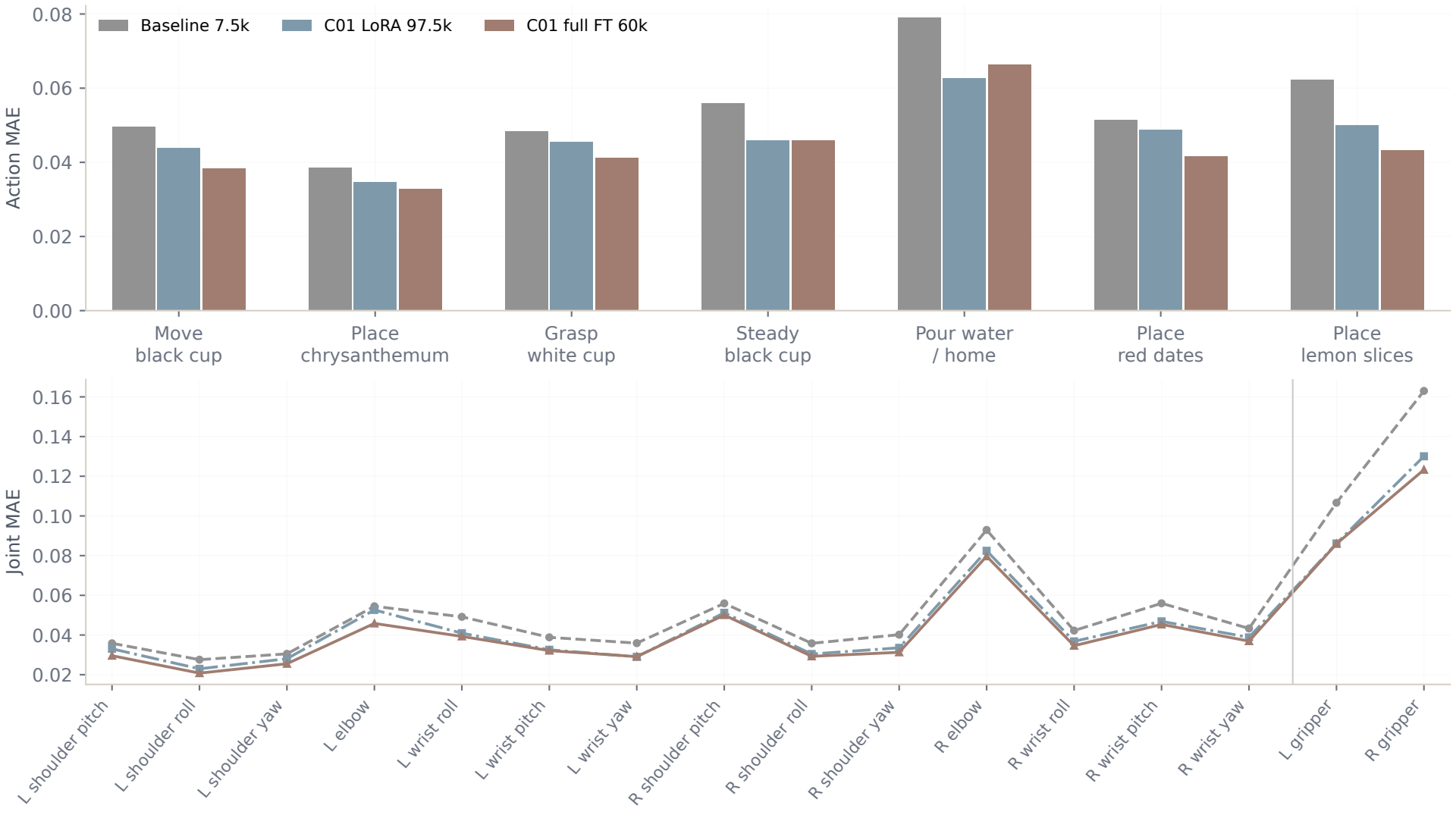
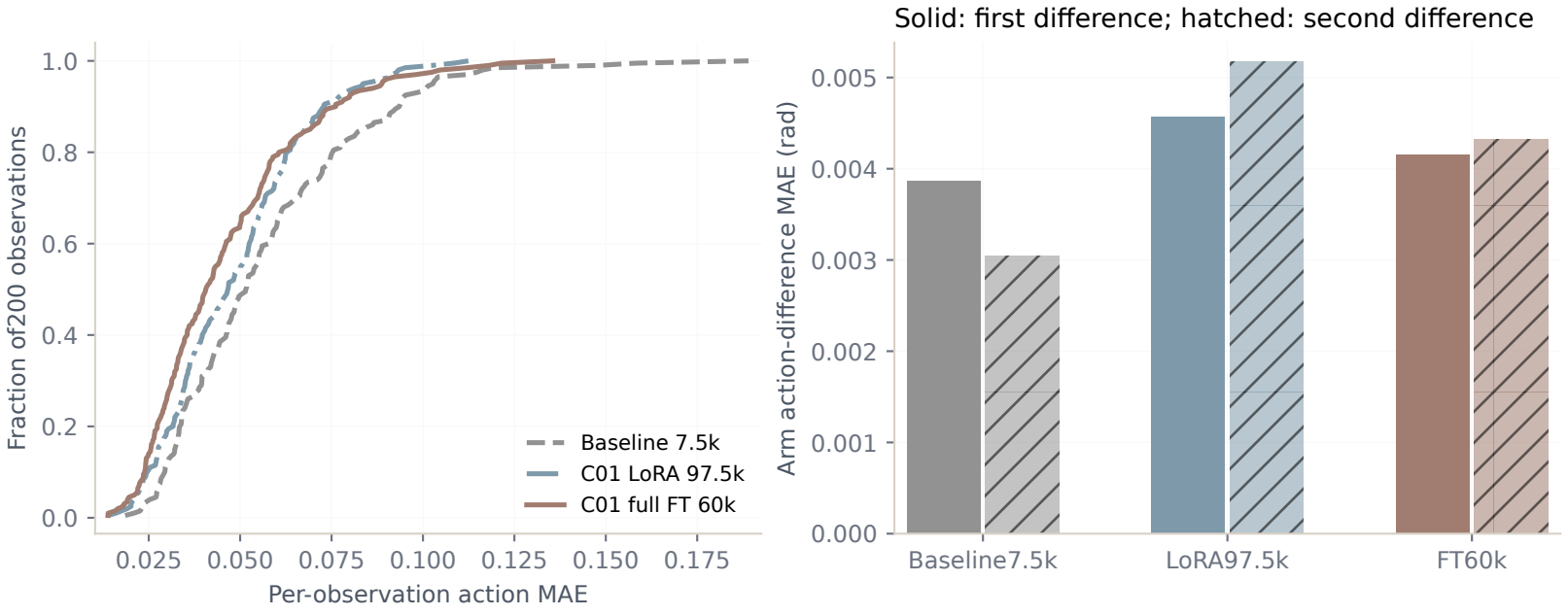
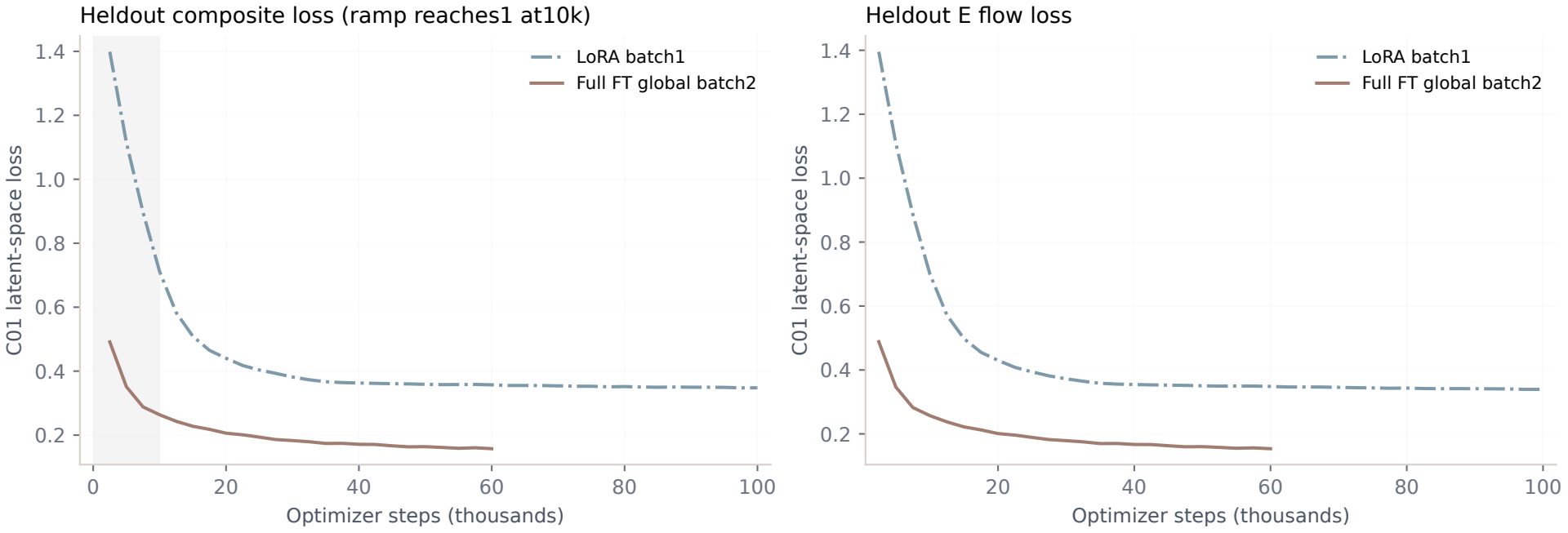
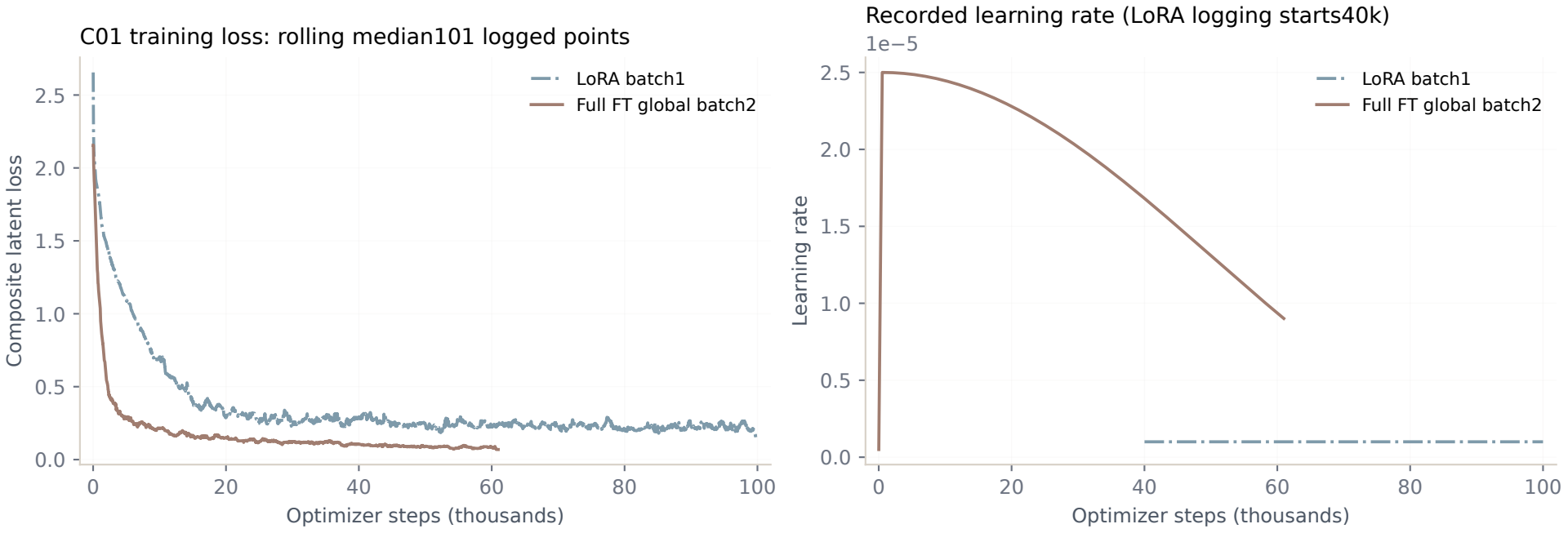

Matched current observations: 200 evaluation rows, 30-step / 16-joint actions
Error bars: 95% cluster bootstrap intervals (20 source episodes)
Action MAE
Native position error
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.0568
0.0485
0.0462
Baseline 7.5k
LoRA 97.5k
Full FT 60k
Action RMSE
Native position error
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.1268
0.1033
0.1205
Baseline 7.5k
LoRA 97.5k
Full FT 60k
Range-normalized MAE
Fraction of training range
0.00
0.01
0.02
0.03
0.04
0.0355
0.0307
0.0291
Baseline 7.5k
LoRA 97.5k
Full FT 60k

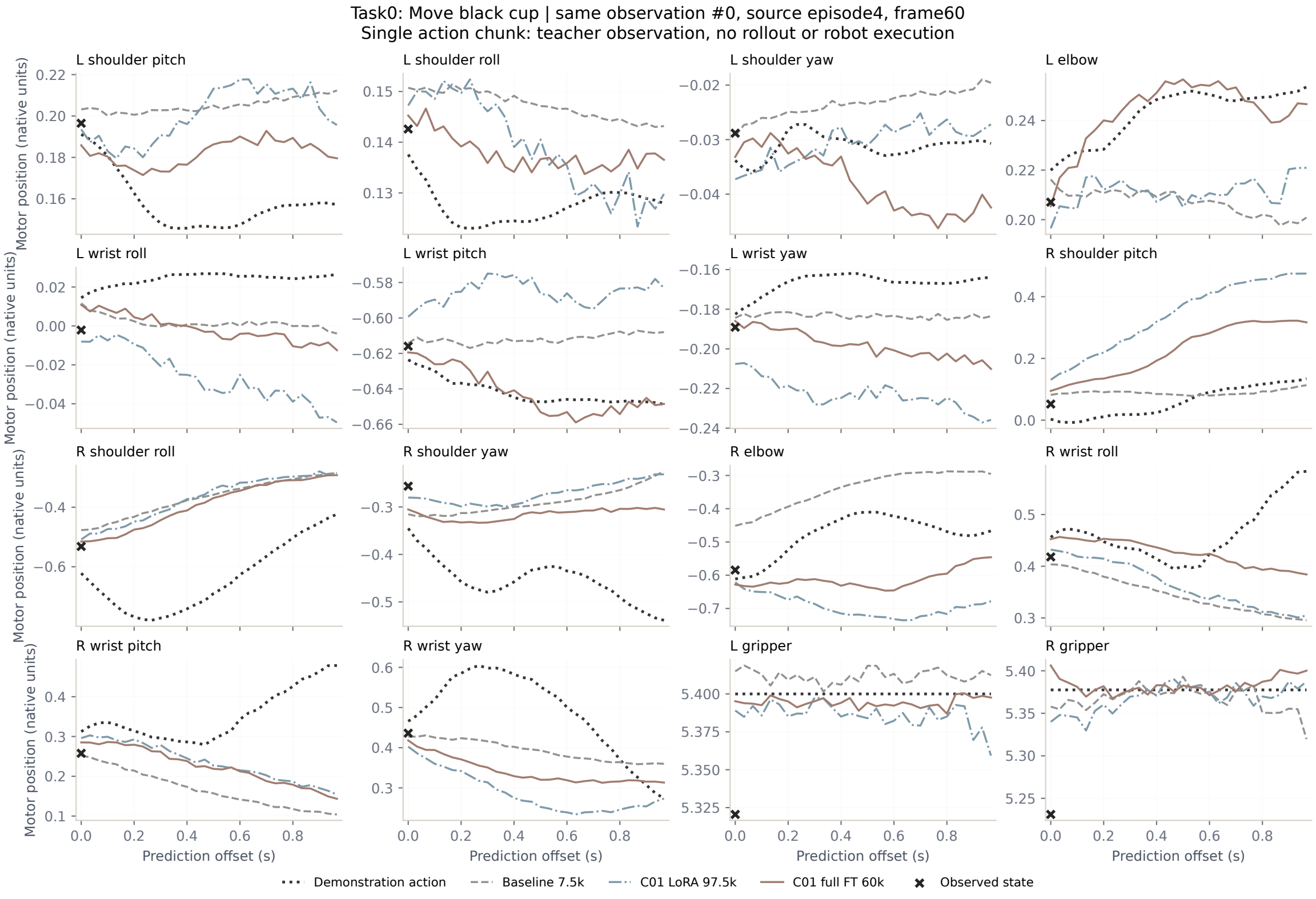

All16 joints
Arm14 joints
Two grippers
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Hold current state
MAE (native units)
Prediction offset (s at30Hz)
0.02
0.04
0.06
0.08
0.10
0.12
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.0
0.2
0.4
0.6
0.8

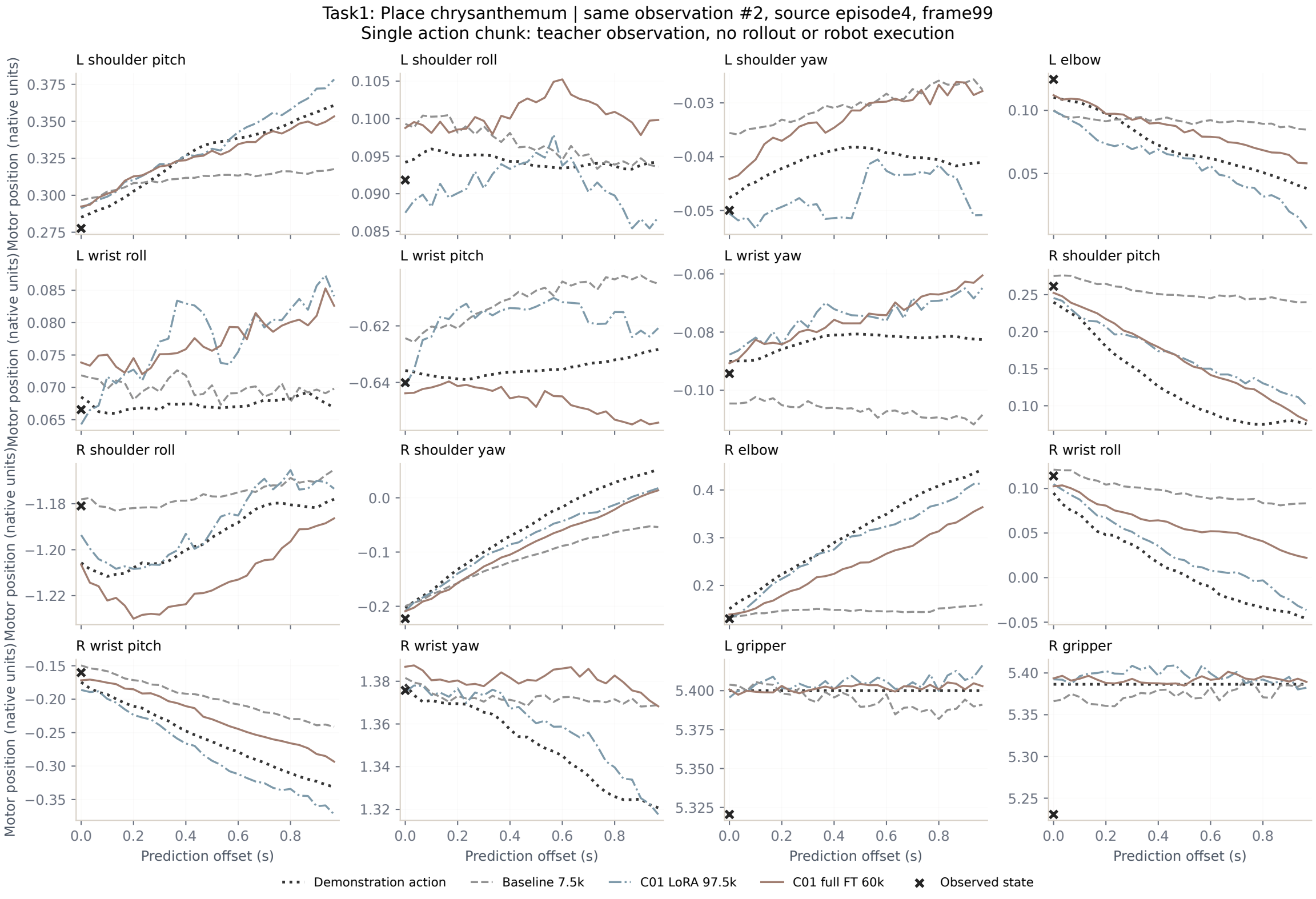

Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Action MAE
0.08
0.06
0.04
0.02
0.00
Move black cup
Place chrysanthemum
Grasp white cup
Steady black cup
Pour water / home
Place red dates
Place lemon slices
Joint MAE
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper

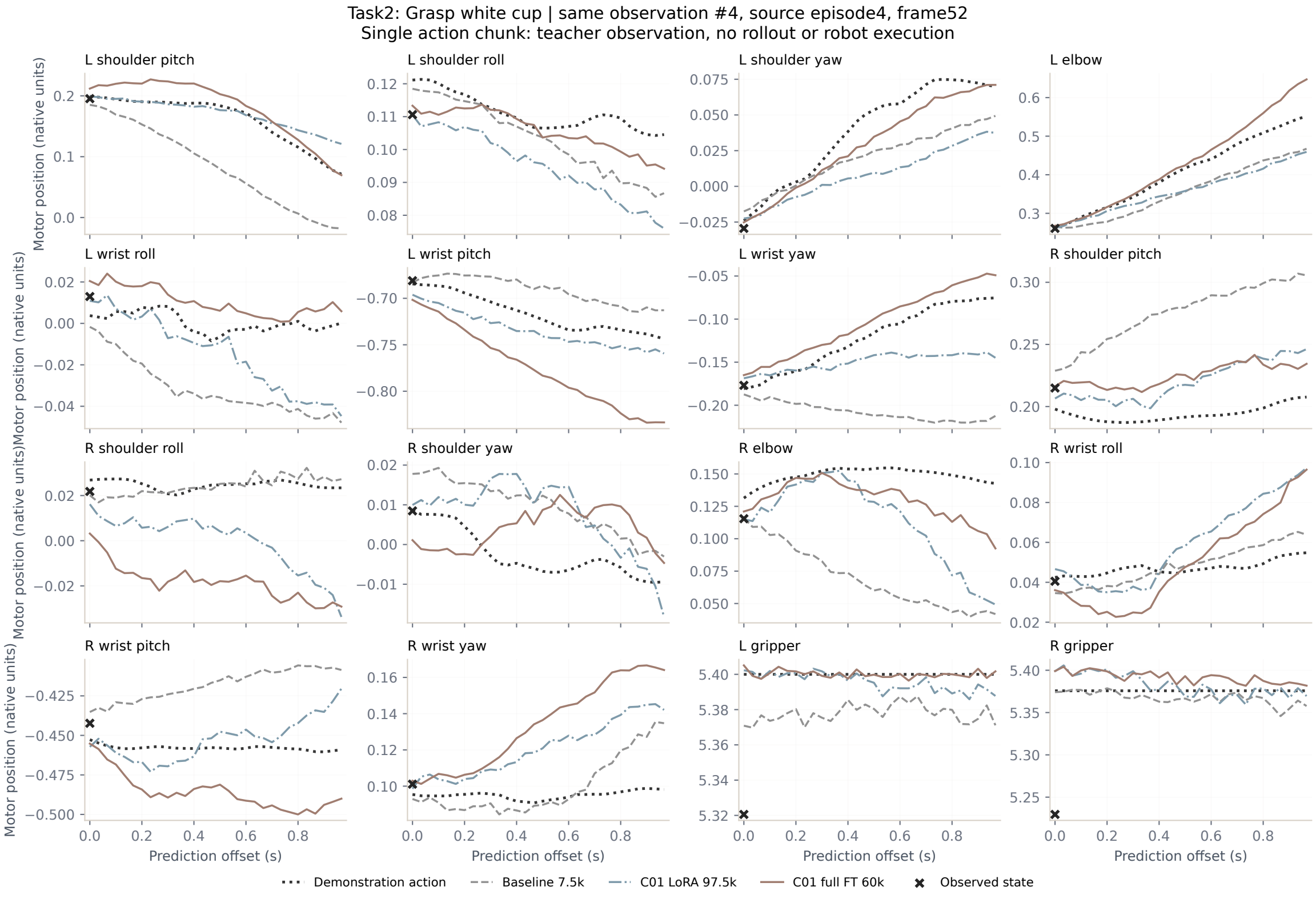

Fraction of200 observations
1.0
0.8
0.6
0.4
0.2
0.0
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Per-observation action MAE
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Solid: first difference; hatched: second difference
Arm action-difference MAE (rad)
0.005
0.004
0.003
0.002
0.001
0.000
Baseline7.5k
LoRA97.5k
FT60k

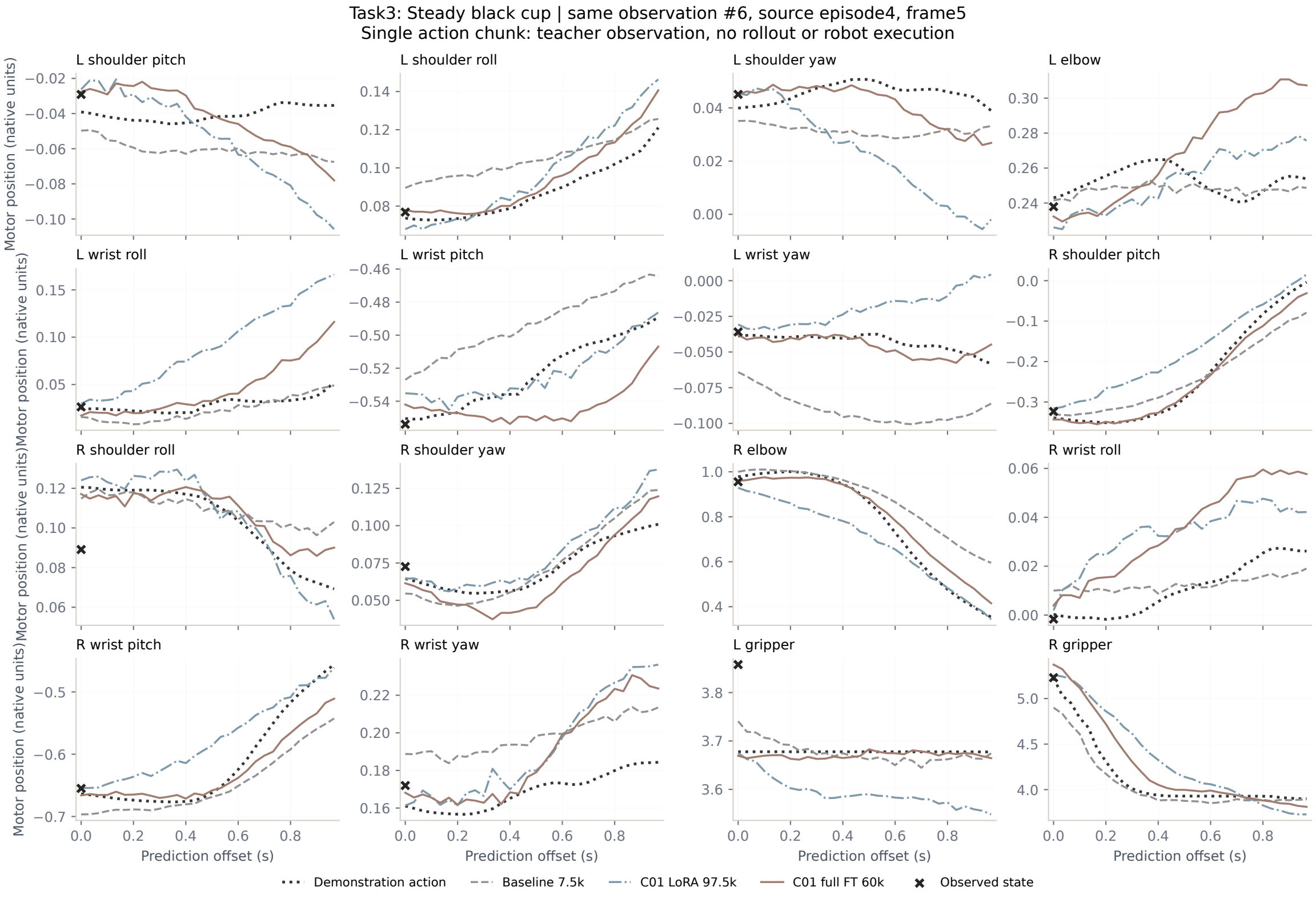

Heldout composite loss (ramp reaches1 at10k)
C01 latent-space loss
Optimizer steps (thousands)
LoRA batch1
Full FT global batch2
Heldout E flow loss
C01 latent-space loss
Optimizer steps (thousands)
LoRA batch1
Full FT global batch2

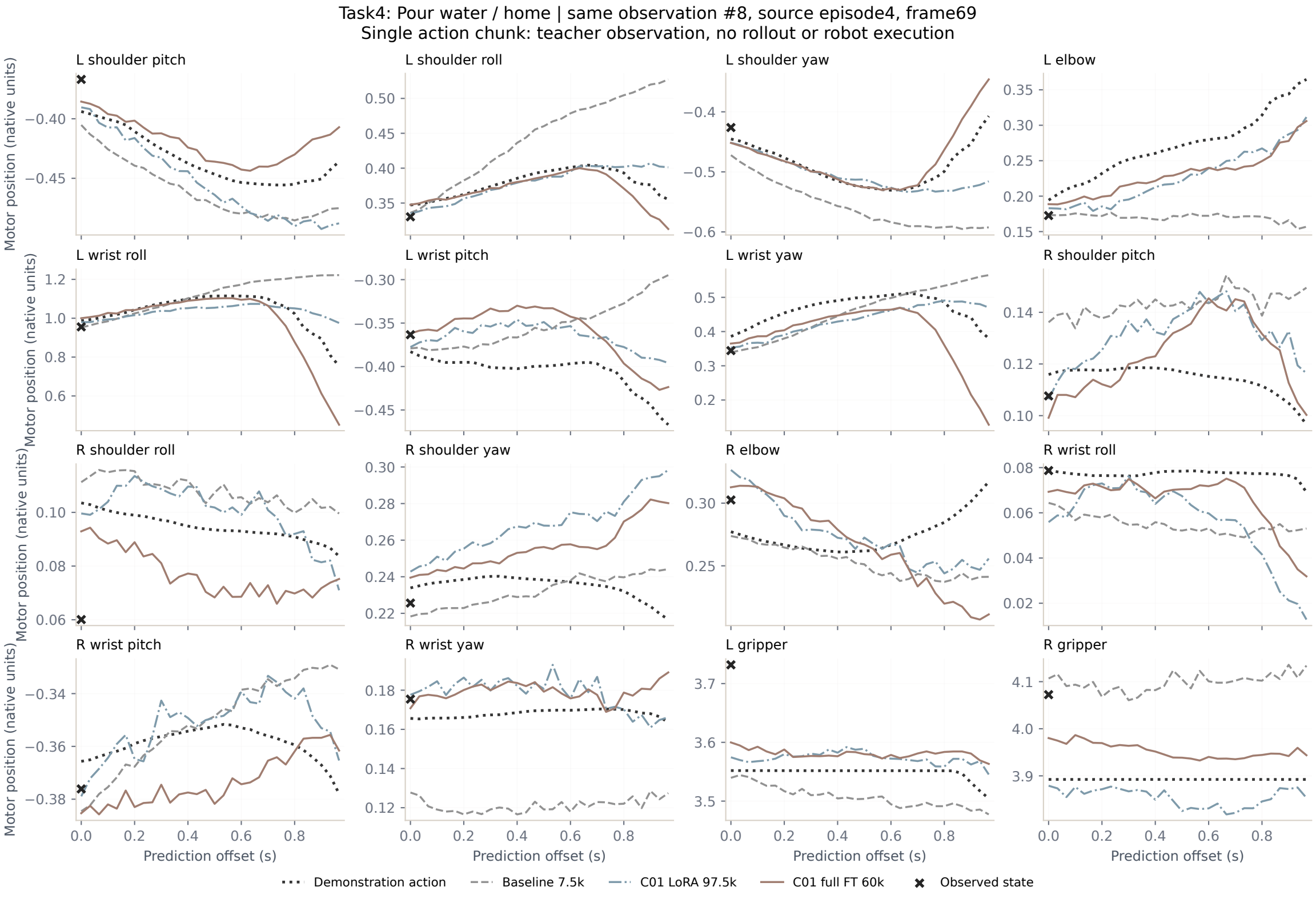

C01 training loss: rolling median101 logged points
LoRA batch1
Full FT global batch2
Composite latent loss
0.0
0.5
1.0
1.5
2.0
2.5
0
20
40
60
80
100
Optimizer steps (thousands)
Recorded learning rate (LoRA logging starts40k)
1e−5
LoRA batch1
Full FT global batch2
Learning rate
0.0
0.5
1.0
1.5
2.0
2.5
0
20
40
60
80
100
Optimizer steps (thousands)

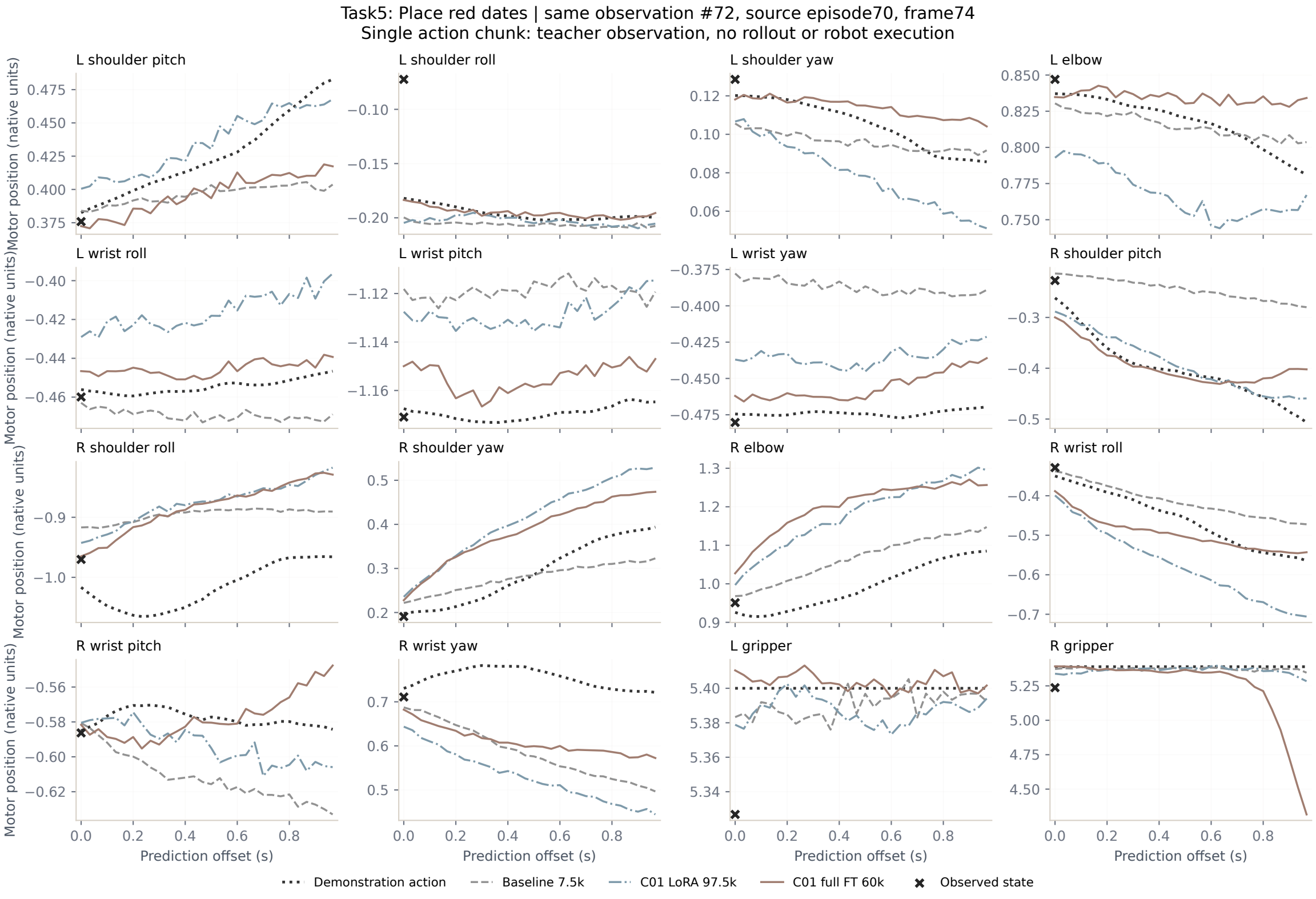

Task0: Move black cup | same observation #0, source episode4, frame60
Single action chunk: teacher observation, no rollout or robot execution
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper
Motor position (native units)
Prediction offset (s)
Demonstration action
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Observed state

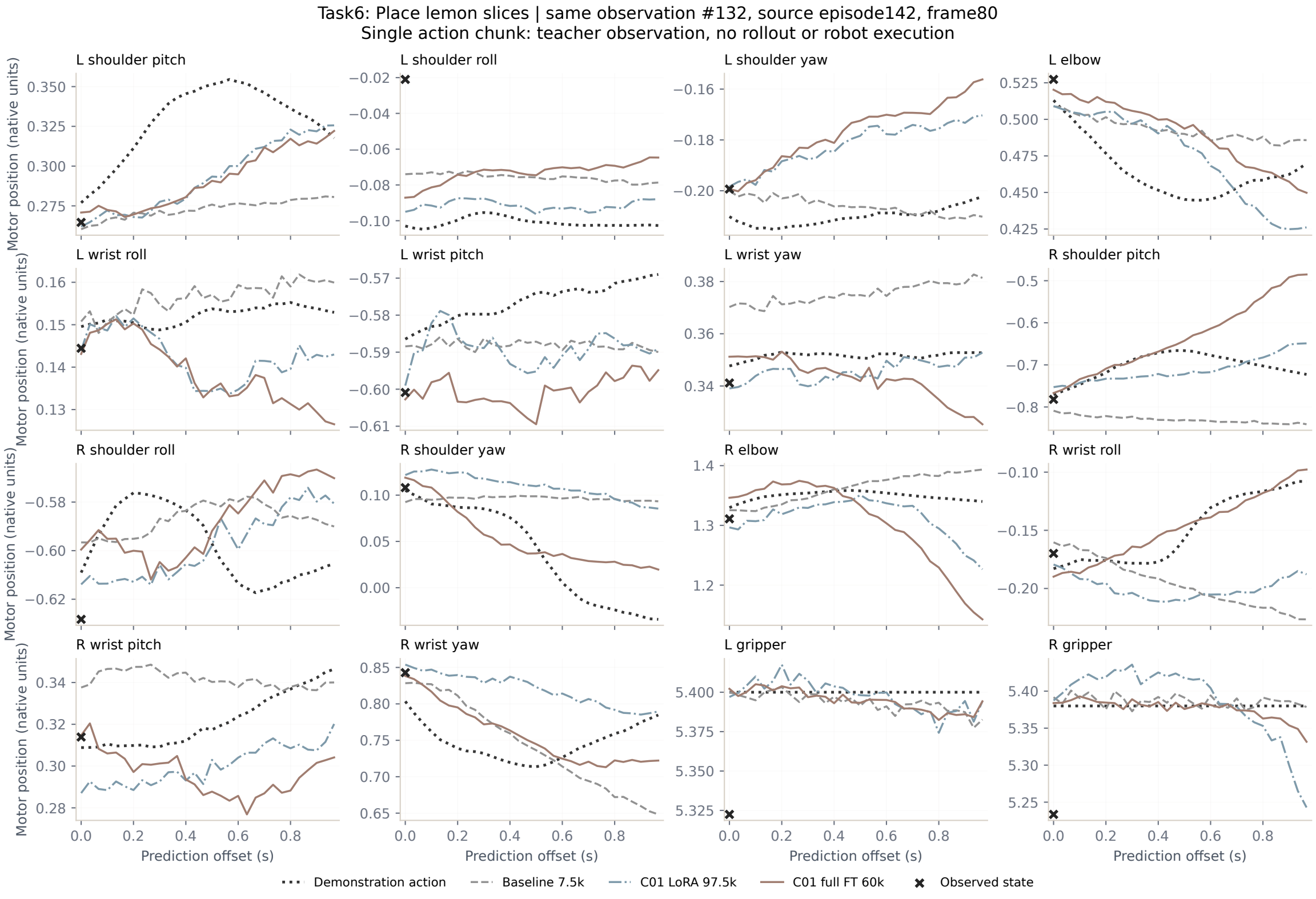

Task1: Place chrysanthemum | same observation #2, source episode4, frame99
Single action chunk: teacher observation, no rollout or robot execution
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper
Motor position (native units)
Prediction offset (s)
Demonstration action
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Observed state


Task2: Grasp white cup | same observation #4, source episode4, frame52
Single action chunk: teacher observation, no rollout or robot execution
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper
Motor position (native units)
Prediction offset (s)
Demonstration action
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Observed state

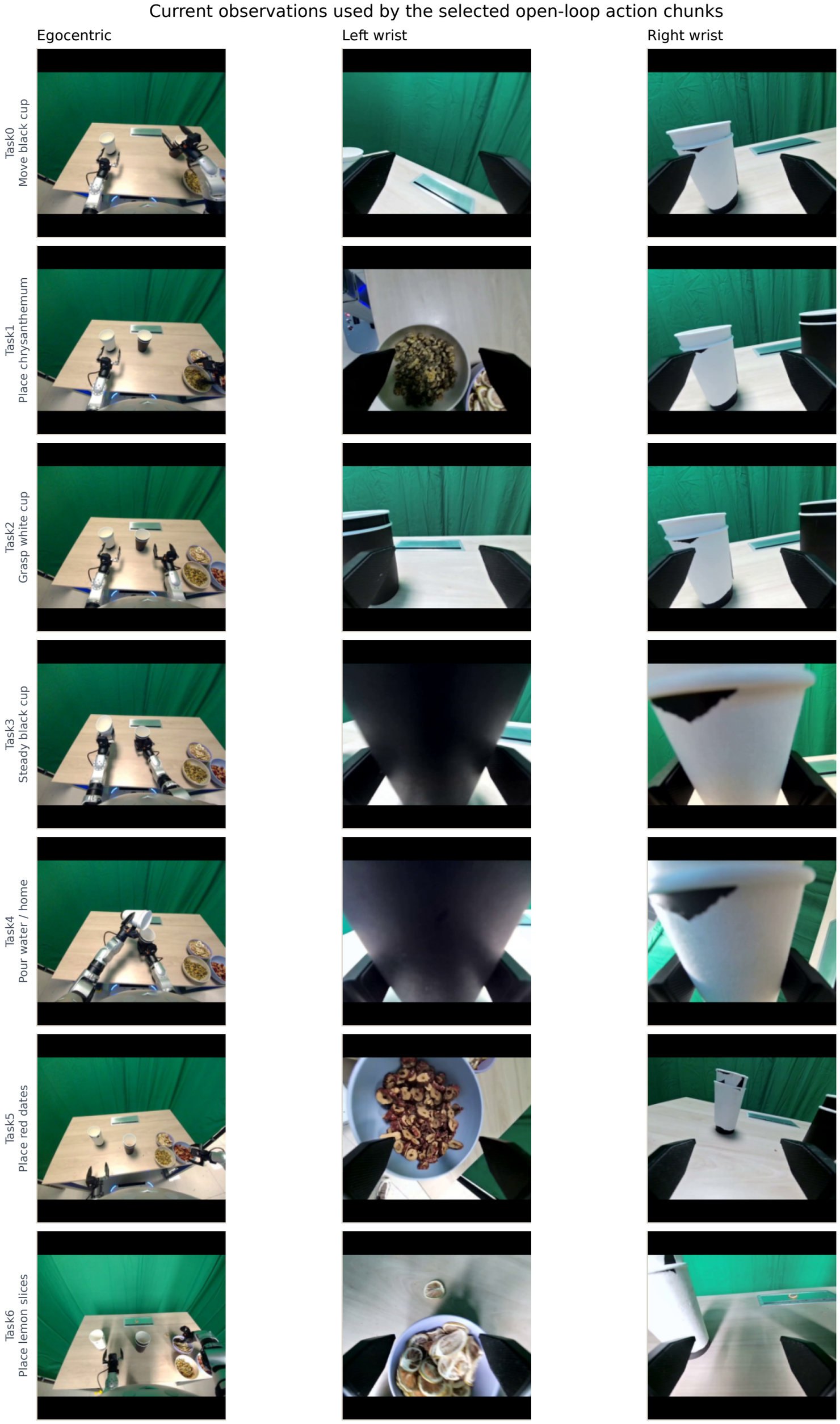

Task3: Steady black cup | same observation #6, source episode4, frame5
Single action chunk: teacher observation, no rollout or robot execution
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper
Motor position (native units)
Prediction offset (s)
Demonstration action
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Observed state

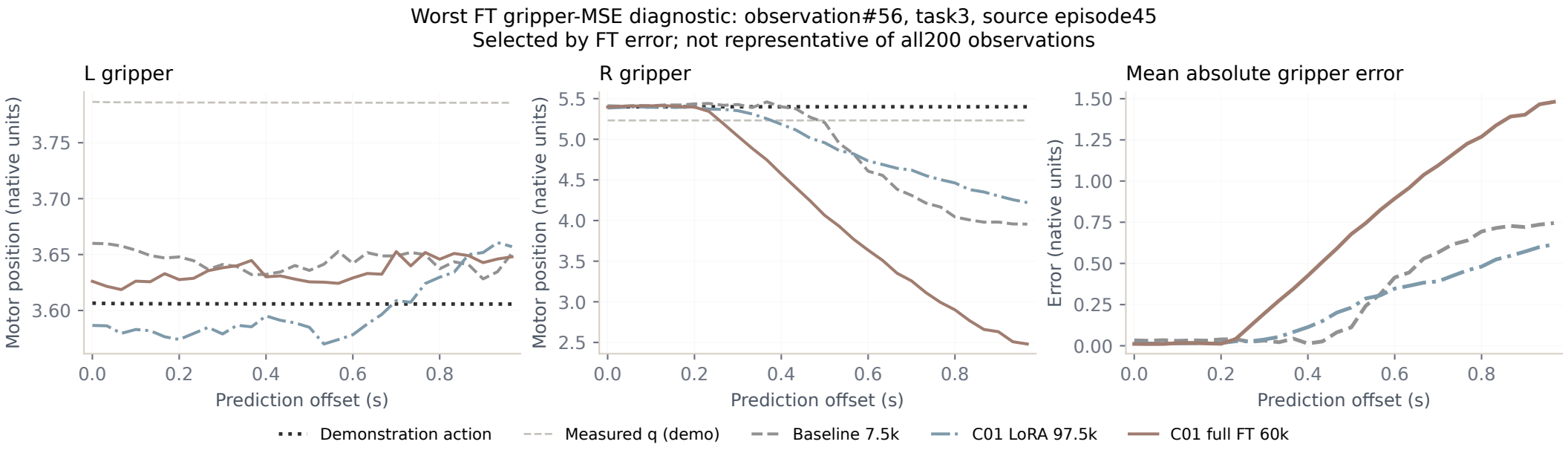

Task4: Pour water / home | same observation #8, source episode4, frame69
Single action chunk: teacher observation, no rollout or robot execution
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper
Motor position (native units)
Prediction offset (s)
Demonstration action
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Observed state

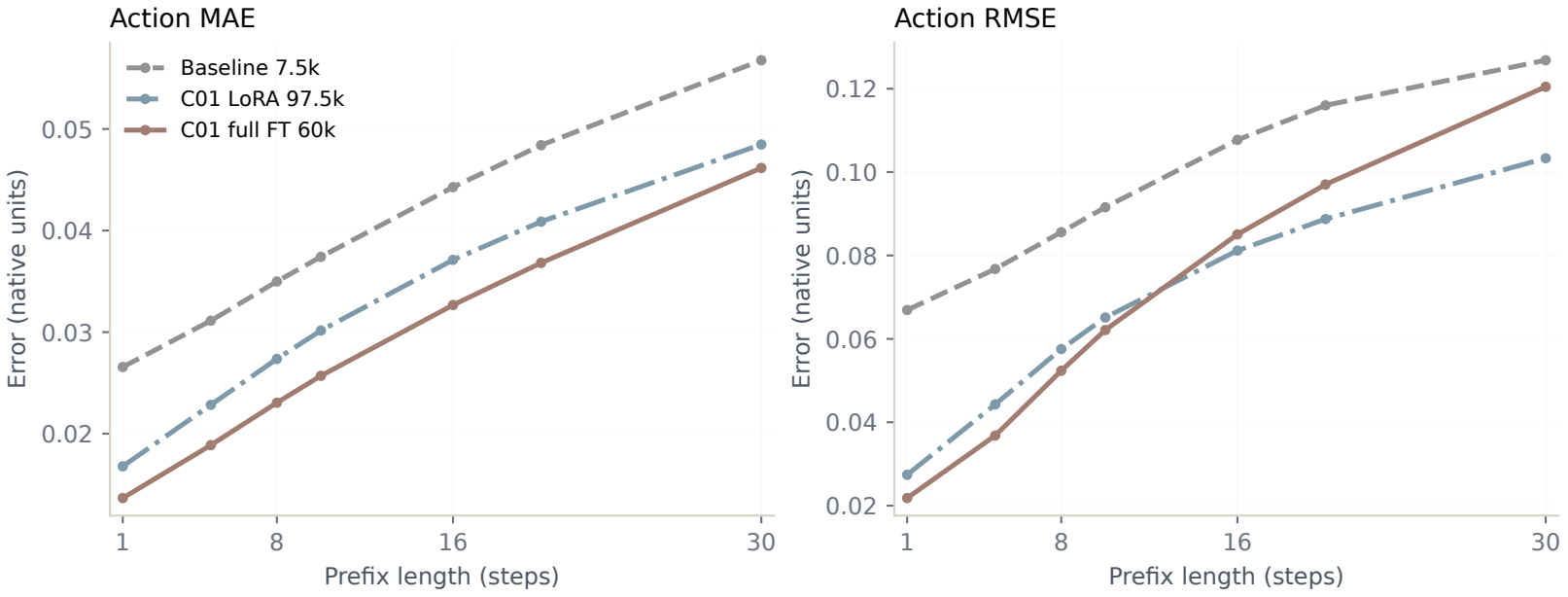

Task5: Place red dates | same observation #72, source episode70, frame74
Single action chunk: teacher observation, no rollout or robot execution
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper
Motor position (native units)
Prediction offset (s)
Demonstration action
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Observed state


Task6: Place lemon slices | same observation #132, source episode142, frame80
Single action chunk: teacher observation, no rollout or robot execution
L shoulder pitch
L shoulder roll
L shoulder yaw
L elbow
L wrist roll
L wrist pitch
L wrist yaw
R shoulder pitch
R shoulder roll
R shoulder yaw
R elbow
R wrist roll
R wrist pitch
R wrist yaw
L gripper
R gripper
Motor position (native units)
Prediction offset (s)
Demonstration action
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Observed state

Current observations used by the selected open-loop action chunks




Worst FT gripper-MSE diagnostic: observation#56, task3, source episode45
Selected by FT error; not representative of all200 observations
L gripper
R gripper
Mean absolute gripper error
Motor position (native units)
Error (native units)
Prediction offset (s)
Demonstration action
Measured q (demo)
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k


Action MAE
Baseline 7.5k
C01 LoRA 97.5k
C01 full FT 60k
Error (native units)
0.05
0.04
0.03
0.02
1
8
16
30
Prefix length (steps)
Action RMSE
Error (native units)
0.12
0.10
0.08
0.06
0.04
0.02
1
8
16
30
Prefix length (steps)